

performer. At that time a reference to the use of the keyboard was inevitable in piano composition, and within those limitations, Brown defined the boundaries of each system through two continuous lines moving from left to right, indicating the outer limits of the keyboard.

Mario Bertoncini's interpretation 15 years later shows what wealth is stored in the extraordinary instructions. It was now possible that Bertoncini could adapt the symbols, movement, and space to his own tonal materials. Similar to other works of his, for example «CIFRE,» he used the flexibility of the grand piano as keyboard instrument, percussion instrument, stringed instrument, and simply as a resonating body.

The long lines in the horizontal indicate the sustained notes, which at the start of the piece are positioned through a hanging cymbal in the sounding board region of the piano, and are played with a violin bow, whereby flageolet-like overtones are produced by a special gripping technique used on the cymbal. Other sounds of this nature are obtained through the movement, for example, of a guiro (in this case, special guiros made from bamboo) along the strings of the piano. The smallest graphic symbols are realized through the striking of the iron frame and the sounding board by the wooden, felt, or rubber mallets of the percussionist while cow-bells and wood-blocks rest on the strings.

Through these sonic materials and with the help of the three channel playback-method, Bertoncini was able to «wander through» the complete symbols of space from left to right, system by system.

The score, divided into three layers, of which each is a partial «filtered version» of the complete marked score, is brought together again through overlapping the single versions so that it can be heard by the audience in its complete musical realization.

Almost 20 years after the recording session, and 35 years after its creation as a composition, this recording documents the strength of E. Brown's piece.

CIFRE

CIFRE has reached the audience of Europe, as well as the concert halls of the United States and Japan in ensembles ranging from one to seven performers. It was composed for a variable number of pianists in which each performer has two «prepared pianos » at his disposition. Building upon the techniques of post - serialism, which adapted the modern piano in the sense of using its traditional sonic possibilities as well as exploiting its percussive qualities, and in the qualities given it through the methods of the «prepared piano» developed by John Cage, Bertoncini achieves a multifaceted, richly variegated palette of colors which corresponds to a purely acoustical nature, which in no respect whatever relies upon electronic manipulation. Special characteristics of the work include the «Klang-bänder,» or sound bands long-sustained notes, *con l'arco*, which stand alone in utter contrast to the innate qualities inherent in the piano's sound, as far as its mode of attack is concerned, but which, however, leave the natural resonance of the instrument intact. In opposition to the rapid aperiodical rhythmic impulses that stand as a sort of «traditional frame» in the economy of the whole work, there lies a quiet, static sound, which evokes the characteristics of a large string ensemble

Thus, in this sense, the instrument has been extended into a third realm, that of the stringed instrument, alongside its qualities as keyboard instrument and percussion instrument.

For Bertoncini these and similarly developed methods were not limited to the importance of an instrumental technique, but had their value in a purely compositional speculative process. He extended these methods to percussion instruments and, as a result, was led to intensive participation in concert activities at the beginning of the 1960's, as well as to his commitment as a member of the improvisation group «Nuova Consonanza di Roma,» the first of its kind, which he joined in 1965,

one year after its formation.

The novelty of the «Nuova Consonanza» group - as Bertoncini in several occasions has pointed out - consisted in a collective «real time» composition, so to speak, in which the performing quality of each individual contribution was totally subdued to the requirements of a pure musical thought: the composer - player was not a performer but instead his action consisted strictly in adding a tone colour to the whole, as an element of a larger mosaic. The meaning of the whole improvisation itself could only be understood through the sum of its parts, which would sound utterly meaningless to the listener when heard alone and out of context.

Only through making such specific distinctions from similar collective improvisation groups (which on the contrary are mainly based upon performing and often virtuoso - like exhibitions) can a historical evaluation of «Nuova Consonanza» be drawn in concrete terms and its sphere of activity clearly determined.

His personal contribution to the collective work can be seen in the results of his methods in sound production in relation to the grand piano and percussion instruments (the use of the bow - hairs of stringed instruments and the materials which evoke CIFRE's long sustained tones), as well as their public promotion through the framework of numerous concerts in Italy and in foreign countries — also in radio and studio recordings made by RCA and Deutsche Grammophon.

The implications of this treatment of the instrument were first described in detail in the published score, which appeared in *Source* magazine (Davis, California) in 1968. The piece and its technique of composition, which was conceived in 1963, but finally scored between 1964 and 1967, was the first of its kind. Because of its rapid popularity among other composers and interpreters of new music, Bertoncini felt forced to distance himself from the usage of such techniques in his subsequent work.

Similar paths taken in composition during this phase of the «Informal» in the arts, could now be at-

tempted through experiments with sound and the development of new sonic resources. It was in that particular atmosphere of the 60's that it was attempted, to thoroughly consequence to emancipate oneself from the traditional elements of music, and from the academic baggage which accompanied a musical language whose morphological elements no longer proved sufficient in transmitting a valid message.

The emancipation, in this case, also affected the formal aspects of composition. In CIFRE particularly, the structure of the piece is inseparably interwoven into the fabric of its sonic elements, and also in regard to the methods of its creation, which are at once unrepeatable and decisively affect not only the formal, but also the stylistic characteristics. Perhaps they form a condition of compositional necessity, if not, at best, the personal signature of the composer.

The «form» of CIFRE, appropriately marked by Bertoncini as «forma-meccanismo» and «forma organismo,» could be considered as an anthropomorphic structure, i.e. a human body with both its capabilities and limitations. Each «organ» is able to perform its specific task, whereby a strict time limitation is not pre-established. The function and the particular use of the individual «parts» are predetermined by the composer. It remains up to the performer, or performers, to use the elements of the sound materials within a defined span of time, and to choose their order of sequence. This lends the piece its richness of variety without, however, disposing with the composer's original intentions.

Observing the score of CIFRE, it becomes apparent that the different musical materials are placed in a sort of Cartesian plane, whose abscisses and ordinates are delineated into time and octaves, respectively, and point to the areas of pitch on the piano where the musical action is to take place. The notation stems from symbols and markings which, while partly indebted to tradition, here however, stand in a completely different context, and are accordingly used in a different sense.

At the beginning of the score, for example, there is a shaded-in crescendo marking which symbolizes the transition of a sustained note from piano to forte. At another place in the score, there are cowbells represented as ovals lying on the strings; at yet another place, movements of the hand through sinus-wavelike figures indicate how the performer is to move over the strings with brushes. For the procedures delegated to the keyboard, there are fingerings given similar to tablature, in which the right hand fingerings receive arabic numerals, the left, roman numerals. It is thus not the actual musical result which is notated, but the coordination of both hands through which the section is abstractly represented. The score is to be read from left to right and through repeated playings takes on the character of a round. The varied musical materials can be seen as sections of this circular process. The choice of segments is left up to the performer, however he must still take into consideration the temporal logic of his path in the piece. In this manner, it becomes impossible to go from one material to the next which «temporally» took place at an earlier «time». If one jumps over certain sections, as it were, these jumps are filled out with rests, as if the unplayed actions were replaced by a corresponding period of time. One such musical segment, once closed, may not be repeated, neither during the playing of the passage an hand, nor in further passages, and also not by other performers should others be involved in the performance. In this respect, it is possible that a time lag and a consequent overlapping of the various sections can occur, which in its musical flow, shifts in many directions. In the event that two musical actions contradict one another, (as for example, in a version for several performers, in which one element is to be played forte and the other piano), one of the performers must interrupt his action and replace it with an equivalent pause.

The versions which are retained on this recording are for a single performer (Mario Bertoncini) and for a group of three (Maura Cova,

Alberto Neuman, Mario Bertoncini). One recognizes the diversity inherent in this piece, and the number of combinations of musical events that are possible. Where is the form? In one of the other versions or is it contained in the ideas themselves which determine the structure in its entirety? This question is equally applicable when discussing Cage's «CARTRIDGE MUSIC» and Brown's «FOUR SYSTEMS.»

CARTRIDGE MUSIC

With the simple phrase, «Let electronic music live,» John Cage ushered in a new epoch of electronic music.

The wish of composers to pursue new tonal pathways unburdened by tradition and the limitations of acoustical instruments and their inherent compositional restrictions, came closer to being fulfilled with the development of electronic equipment after World War II. *Musique concrète*, still bound by its reliance on natural sound sources, yielded quickly to the advancements of electronic oscillators, with which sound, now produced synthetically, was placed in complete control of the composer himself. The long dreamed of goal of subordinating the parameters of music to totally serialized techniques, and above all, keeping it under the sole control of the composer, seemed to have been achieved. The restrictions involved with the arbitrary manipulation and the extension of sound through acoustical instruments, originally conceived with a different musical purpose in mind, would no longer be applicable because of the development of new technical possibilities. The failings of the performer would not be acceptable in a world where the performer is no longer capable of meeting the demands placed on him and his instrument, and is not able to cope with the growing rhythmic complexities of new music. It was only now that the composer thought he could, in accordance with the amount of work involved in electronic studios under supervision from technicians, to mould his work until the result of his expectations was fulfilled. Similar to the visual artist, he sought to deliver his art directly to his audience without having to rely upon the imperfect uncontrollable manipulations through an intermediary human being, the performer. Every aspect of a composition could be harnessed to his will — even the risks involved in the performance itself.

The toilsome labor, which, until then formed the basis of electronic music was simplified during the

course of the 1960's as a result of the work done by engineers, mainly through the work of A. Moog. Many processes which otherwise could only have been carried out manually, were shortened considerably as a result of the automation of the voltage-control-systems. The illusion of imitating a real musical phenomenon, such as the bow of the violin striking a string, (simultaneously influencing several parameters, nevertheless, voluntarily being able to modify them) was strengthened by the new technical developments. Despite the enormous progress, it is well known that these technical advancements were far from providing a satisfactory equivalent to manually produced sounds. Even before this development, Cage's CARTRIDGE MUSIC of 1960 made the attempt to bring together the fallacies of the human being with the medium of electroacoustics, not as a necessary evil, but as a structural element of his composition. The listeners, and/or spectators, become witness to the creation of sonic events which are unavoidably attached to the theatrical quality of the performer's actions, who is finally able to subtly control his electroacoustical environment. The arrangement of cartridges and contact-microphones are individually connected to amplifiers and loudspeakers and individually operated by the performers. The listeners sees, as it were, through a magnifying glass, which suddenly opens up the strange world of sounds and acoustical occurrences which otherwise would remain hidden from view. A world which doesn't coincide with traditional or electronic music (and which was thought of as undesirable) is brought into the foreground, in which «auxiliary sounds,» like the sound of objects on a table revealed by contact microphones, become audible for the first time. Even the former hated «humming» of an electronic amplifier is «welcomed» under such circumstances. The picture of an impish John Cage grinning as he hunts mushrooms comes to mind. The main sonic events result from objects attached inside phono cartridges (former pick-ups from phonographs) where the needles were once located. These

elements form a versatile and variegated fabric of sounds and noises possessive of their own unusual poetry.

In his suggestions for the choice of objects to be used, Cage has a clear spectrum of sounds in mind. For example, a piece of wire, a pipe-cleaner, or a Slinky (a metal spring, which upon being touched produces, through its up and down movements, especially resonant sounds) can be incorporated into the performance. The performer retains the right, however, to bring into play his own choice of objects, in order to discover and illuminate an added facet in the realm of the sonic events. Bertoncini, using other objects (in mentioning the slinky), was prompted to incorporate different springs from the works of clocks into the performance, which produce a similar resonating effect, and make for a humorous contradiction between the visual point of reference and the phonic result. An extremely small clock spring, for example, produces a deep sound.

CARTRIDGE MUSIC's form arises out of the charm of the limited chance and from the struggle between the willed action of the performer and the imagination of the composer.

The score consists of several «layers,» positioned according to a specific abstract graphic form. The performer chooses between 20 pages, each of which is printed with one or more (1-20) irregular fields representing the cartridges. Four different transparent sheets are placed over a chosen page which also displays different forms and graphic symbols. One sheet, marked with dotted lines, indicates the musical time brackets during the course of the entire action, another, with accidentally ordered dots, is marked with small circles, as well as a larger circle, in which the outlines of a stopwatch are engraved. If during a pass through in this fashion, a line of one of the fields which designates the cartridges lands on one of the dots in the corresponding underlying sheets, this means a tone is to be created by means of one of the objects mounted in the cartridges. Upon contact with a circle, the change of sound colour is

affected through the knob of the amplifier's regulator. The method of ordering the «layers,» or transparencies, determines the single actions of the performer, who is left the possibility to carry out the order in such a manner that his intentions come as close as possible to being realized. Consequently, however, the forms prescribed by Cage are the only ones at one's disposal, and the accompanying interpretation which occurs through the layers does not necessarily lead to any given result. During a performance with several performers, the results are often an interplay, which under certain circumstances can lead to a cancelling out, or, in the opposite sense, an exponential expansion, due to the combinations of the particular performer's scores.

The meaning behind Cage's aleatoric processes of composition lies in the emancipation of the antipathies and preferences of the performing participants of the musical process — as he remarked in an interview with Hans G. Helms — the widening of the horizon for previously unknown sonic events.

Having chosen an arrangement of his score, the performer is bound by this particular course of action until it is brought to a conclusion. At the same time as Cage preorders his boundaries of the musical events, he always takes into consideration their potential character.

«And were I to go to China I would, I believe, only be useful to the Chinese as a mushroom hunter, but I might also be useful as a person who could point out the pleasure of environmental sound which I think is accessible — well, as we say here in the United States — to the man in the street.» (Cage in an interview with Hans G. Helms in New York, 1972)

Martina Schaak
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Translated by David Walters.

Mario Bertoncini

1932: born in Rome
1951-'61: studies in classics and music (compositions with G. Petrassi and piano with R. Caporali) in Rome, electroacoustic music in Utrecht (G. M. Koenig).
1962: experiments with prepared instruments (Quodlibet, Cifre, Tune, Scratch-a-matic).
1965-'72: member of the improvisation group «Nuova Consonanza» in Rome.
1967: music theater. Theoretic and practical work on a functional interaction of sound and gestures («Spazio Tempo», «Note per un teatro della realtà».)
1968: «Epitaffio»: correspondence between the action of drawing and writing a text and its sonorous result.
1970: première (Venice, Biennale d'Arte) of «Spazio-Tempo», audio-visual piece for dancers/mimes and four instrumental groups. Construction of his first sound objects (principle of the Aeolian harp).
1975-'76: professorship at «McGill University», Montreal: «Musical Design Course» (founder of the group, later known as «SONDE».)
1986: invention of the «Choreophon» (transformation of dance-gestures in optical and acoustical signals. Notification of german patent DE 36 43 018 A 1). Co-founded the theater group «VIE» with Martina Schaak and Roberto Capanna.
Lives and works in Berlin.

Maura Cova

After completing her piano degree in Rome (S. Cecilia) under the supervision of Vera Cobbi Belcredi, participated in Arturo Benedetti-Michelangeli's Master Classes, «Corsi di Perfezionamento pianistico», in Bozen, Arezzo and Bergamo, Italy.
Attended the counterpoint and analysis courses of Antonio Ferdinandi in Rome.
Awarded a diploma as «Artist Therapist» from the «American Association of Artist-Therapists», in New York, USA.
Co-founded the «Center for Experimental Music Studies» in Rome with A. Neuman.
Therapeutic experiments (Dora Kalff's methods) with children concerning the interrelation between music and drawing.
As of 1983, developer along with John Heineman and Eugenio Cavallari of a Personal Computer System for handicapped children called SCALE.
Continued teaching activities in Rome at the «Istituto Magistrale», «Scuola di Danza Astaldi», «Liceo Sperimentale», and the «Centre Provençal d'Animation» in Marseille.
Use of the Personal Computer (Macintosh) for musical pedagogical purposes. Concert appearances as soloist in Italy and in foreign countries.
Radio and television recordings of largely contemporary music.

Alberto Neuman

Born in Buenos Aires.
After an early graduation from his piano studies resettled in Italy.
Was awarded a degree of «Curso di Perfezionamento pianistico» under the auspice of Carlo Zecchi at the Academy of S. Cecilia in Rome.
Afterward attended Arturo Benedetti-Michelangeli's Master Classes («Accademia di Interpretazione pianistica») in Arezzo, Italy.
Awarded professorship at the University of Rome.
Co-founder along with Maura Cova of the «Center for Experimental Studies» in Rome.
Teaches piano on the faculty of the «Conservatoire National de Région» and the «Conservatoire Européen» in Paris.
Appointed legal consultant on the committee of author's rights in Paris.
His work as pianist shows his devotion to both the classical and experimental repertoire.
Appearances in solo recitals and as soloist with orchestras in radio broadcasts and television programs in Europe and America.
Records for the Arion and Ades labels in Paris.